

Discrete Mathematics

Week 2: Biconditional, XOR, Simplification & Quantifiers

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Biconditional ($\leftrightarrow$)

Definition

$p \leftrightarrow q$ means “ p if and only if q ” (also written as “ p iff q ”)

This means: $p \rightarrow q$ **AND** $q \rightarrow p$

Symbols: $\leftrightarrow$ or $\Leftrightarrow$

Truth Table:

p	q	$p \leftrightarrow q$
1	1	1
1	0	0
0	1	0
0	0	1

Note: True only when p and q have the **same** truth value!

Properties of Biconditional ($\leftrightarrow$)

Key Properties:

- ① $p \leftrightarrow p = 1$ (Always true)
- ② $p \leftrightarrow \neg p = 0$ (Always false)
- ③ $p \leftrightarrow 1 = p$
- ④ $p \leftrightarrow 0 = \neg p$

Example:

- $p: x > 5$
- $q: x \geq 6$ (for integers)
- $p \leftrightarrow q$: “ $x > 5$ if and only if $x \geq 6$ ”
- For integers, these statements are equivalent!

Biconditional - Computer Science Example

Example in Programming:

“A function is recursive if and only if it calls itself.”

$$p \leftrightarrow q$$

- p : Function is recursive
- q : Function calls itself

Truth Table Analysis:

Is Recursive	Calls Itself	Statement
Yes	Yes	True
Yes	No	False (Contradiction!)
No	Yes	False (Contradiction!)
No	No	True

Exclusive OR (XOR) - $\oplus$

Definition

$p \oplus q$ means “ p exclusive or q ”

True when **exactly one** of p or q is true, but **not both**.

Symbols: $\oplus$ or $\underline{\vee}$

Truth Table:

p	q	$p \oplus q$
1	1	0
1	0	1
0	1	1
0	0	0

Note: True when inputs are **different**!

Properties of XOR ($\oplus$)

Key Properties:

- ① $p \oplus p = 0$ (Always false)
- ② $p \oplus \neg p = 1$ (Always true)
- ③ $p \oplus 0 = p$ (Identity)
- ④ $p \oplus 1 = \neg p$ (Negation)
- ⑤ $p \oplus q = q \oplus p$ (Commutative)

Alternative Representation:

$$p \oplus q = (p \wedge \neg q) \vee (\neg p \wedge q)$$

$$p \oplus q = (p \vee q) \wedge \neg(p \wedge q)$$

XOR - Computer Science Applications

Example 1: Access Control

- p : User is admin
- q : User is guest
- $p \oplus q$: User must be **either** admin **or** guest, not both!

Example 2: Bit Operations

- XOR is fundamental in cryptography and error detection
- $1010 \oplus 1100 = 0110$
- Self-inverse property: $a \oplus b \oplus b = a$ (useful for encryption!)

Example 3: Toggle Operations

- Toggling a bit: $\text{bit} = \text{bit} \text{ XOR } 1$

Summary: All Logical Operators

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \oplus q$
1	1	1	1	1	1	0
1	0	0	1	0	0	1
0	1	0	1	1	0	1
0	0	0	0	1	1	0

Quick Memory Tips:

- AND: All must be true
- OR: At least one true
- Implication: Only false when promise broken
- Biconditional: Same values
- XOR: Different values

Simplification of Logical Expressions

Goal: Reduce complex logical expressions to simpler forms using:

- Logical equivalences
- Properties of operators
- De Morgan's laws
- Distribution laws

Why Simplify?

- Easier to understand
- More efficient circuit design
- Optimize boolean expressions in code
- Reduce computational complexity

Let's practice with examples from beginner to advanced level!

Simplification - Beginner Examples

Example 1: Simplify $p \vee (q \wedge p')$

$$\begin{aligned} p \vee (q \wedge p') &= p \vee q \vee p' \quad (\text{Distribution}) \\ &= p \vee p' \vee q \quad (\text{Commutative}) \\ &= 1 \vee q \quad (p \vee p' = 1) \\ &= 1 \quad (\text{Domination}) \end{aligned}$$

Example 2: Simplify $p \wedge (q \wedge p')$

$$\begin{aligned} p \wedge (q \wedge p') &= p \wedge p' \wedge q \quad (\text{Associative \& Commutative}) \\ &= 0 \wedge q \quad (p \wedge p' = 0) \\ &= 0 \quad (\text{Domination}) \end{aligned}$$

Simplification - More Beginner Examples

Example 3: Simplify $p \rightarrow (p \vee q)$

$$\begin{aligned} p \rightarrow (p \vee q) &= \neg p \vee (p \vee q) \quad (\text{Implication: } p \rightarrow q = \neg p \vee q) \\ &= \neg p \vee p \vee q \quad (\text{Associative}) \\ &= 1 \vee q \quad (\neg p \vee p = 1) \\ &= 1 \quad (\text{Domination}) \end{aligned}$$

Example 4: Simplify $(p \wedge q) \rightarrow q$

$$\begin{aligned} (p \wedge q) \rightarrow q &= \neg(p \wedge q) \vee q \\ &= (\neg p \vee \neg q) \vee q \quad (\text{De Morgan}) \\ &= \neg p \vee (\neg q \vee q) \\ &= \neg p \vee 1 = 1 \quad (\text{Tautology}) \end{aligned}$$

Simplification - Intermediate Examples

Example 5: Simplify $(p \vee q) \wedge (p \vee q')$

$$\begin{aligned}(p \vee q) \wedge (p \vee q') &= p \vee (q \wedge q') \quad (\text{Distribution}) \\ &= p \vee 0 \quad (q \wedge q' = 0) \\ &= p \quad (\text{Identity})\end{aligned}$$

Example 6: Simplify $(p \wedge q) \vee (p \wedge q')$

$$\begin{aligned}(p \wedge q) \vee (p \wedge q') &= p \wedge (q \vee q') \quad (\text{Distribution}) \\ &= p \wedge 1 \quad (q \vee q' = 1) \\ &= p \quad (\text{Identity})\end{aligned}$$

Simplification - More Intermediate Examples

Example 7: Simplify $[p \vee (p \wedge q)]'$

$$\begin{aligned}[p \vee (p \wedge q)]' &= [p \vee (p \wedge q)]' \\ &= p' \wedge (p \wedge q)' \quad (\text{De Morgan}) \\ &= p' \wedge (p' \vee q') \quad (\text{De Morgan}) \\ &= (p' \wedge p') \vee (p' \wedge q') \quad (\text{Distribution}) \\ &= p' \vee (p' \wedge q') \\ &= p' \wedge (1 \vee q') \quad (\text{Factor out } p') \\ &= p' \wedge 1 = p'\end{aligned}$$

Alternative: Use absorption law: $p \vee (p \wedge q) = p$, so $[p]' = p'$

Simplification - Advanced Example

Example 8: Simplify $(p \rightarrow q) \wedge (p \wedge \neg q)'$

$$\begin{aligned} & (p \rightarrow q) \wedge (p \wedge \neg q)' \\ &= (\neg p \vee q) \wedge (p \wedge \neg q)' \quad (\text{Step 1: Convert implication}) \\ &= (\neg p \vee q) \wedge (\neg p \vee \neg(\neg q)) \quad (\text{Step 2: De Morgan}) \\ &= (\neg p \vee q) \wedge (\neg p \vee q) \quad (\text{Step 3: Double negation}) \\ &= \neg p \vee q \quad (\text{Step 4: Idempotent}) \end{aligned}$$

Which simplifies to:

$$\boxed{p \rightarrow q}$$

The expression is already in its simplest form as an implication!

Practice: Simplification Challenge

Try this yourself!

Simplify the following expression:

$$(p \vee q) \wedge (p' \wedge q')$$

Hint: Use De Morgan's laws first!

We'll show the solution on the next slide...

Practice Solution

Simplify: $(p \vee q) \wedge (p' \wedge q')$

$$\begin{aligned}(p \vee q) \wedge (p' \wedge q') &= (p \vee q) \wedge (p \vee q)' \quad (\text{De Morgan}) \\ &= 0 \quad (X \wedge X' = 0)\end{aligned}$$

Result: This is a **contradiction** (always false)!

Logical Interpretation:

- $(p \vee q)$: At least one is true
- $(p' \wedge q')$: Both are false
- These cannot both be true at the same time!

Tautology and Contradiction

Definitions

- A **tautology** is a statement whose truth value is **always true**
- A **contradiction** is a statement whose truth value is **always false**

Examples of Tautologies:

- $p \vee \neg p = 1$ (Law of excluded middle)
- $\neg(p \wedge \neg p) = 1$
- $p \rightarrow p = 1$

Examples of Contradictions:

- $p \wedge \neg p = 0$ (Law of contradiction)
- $p \leftrightarrow \neg p = 0$

Contingency Statements

Not all statements are tautologies or contradictions!

A **contingency** is a statement that can be either true or false depending on the truth values of its variables.

Example: $p \wedge q$

p	q	$p \wedge q$
1	1	1
1	0	0
0	1	0
0	0	0

This is **neither** a tautology **nor** a contradiction - it's contingent on the values of p and q .

How to Determine: Tautology or Contradiction?

Method 1: Simplification

- Simplify the expression
- If result = 1 $\rightarrow$ Tautology
- If result = 0 $\rightarrow$ Contradiction

Method 2: Truth Table

- Construct complete truth table
- If all entries = 1 $\rightarrow$ Tautology
- If all entries = 0 $\rightarrow$ Contradiction
- Otherwise $\rightarrow$ Contingency

Let's see examples using both methods!

Example: Using Truth Table

Determine if $(p \wedge q) \vee (p \wedge q')$ is a tautology or contradiction.

p	q	$p \wedge q$	$p \wedge q'$	$(p \wedge q) \vee (p \wedge q')$
1	1	1	0	1
1	0	0	1	1
0	1	0	0	0
0	0	0	0	0

Result: This is a **contingency** (neither tautology nor contradiction).

Note: We can simplify this to p using distribution:

$$p \wedge (q \vee q') = p \wedge 1 = p$$

Example: Using Simplification

Determine if $p \rightarrow (p \vee q)$ is a tautology.

Simplification:

$$\begin{aligned} p \rightarrow (p \vee q) &= \neg p \vee (p \vee q) \\ &= \neg p \vee p \vee q \\ &= 1 \vee q \\ &= 1 \end{aligned}$$

Result: This is a **tautology**!

This makes logical sense: “If p is true, then at least one of p or q is true” is always valid!

Definition

Quantifiers are symbols that specify the quantity of elements for which a predicate is true.

Two Main Quantifiers:

- 1 **Universal Quantifier:** $\forall$ (“for all”)
 - $\forall x \in D, P(x)$ means $P(x)$ is true for **all** x in domain D
- 2 **Existential Quantifier:** $\exists$ (“there exists”)
 - $\exists x \in D, P(x)$ means $P(x)$ is true for **at least one** x in domain D

These are fundamental in mathematical logic and formal proofs!

Universal Quantifier ($\forall$)

Notation: $\forall x \in D, P(x)$

Reads as: "For all x in domain D , $P(x)$ is true"

Examples:

1. $\forall x \in \mathbb{R}, x^2 \geq 0$

- Truth value: **True (1)**
- The square of any real number is non-negative

2. $\forall x \in \mathbb{N}, x > 5$

- Truth value: **False (0)**
- Counterexample: $x = 3$ (which is in $\mathbb{N}$ but not > 5)

Existential Quantifier ($\exists$)

Notation: $\exists x \in D, P(x)$

Reads as: “There exists x in domain D such that $P(x)$ is true”

Examples:

1. $\exists x \in \mathbb{R}, 2x - 3 = 7$

- Truth value: **True (1)**
- $x = 5$ satisfies the equation

2. $\exists x \in \mathbb{Z}, x < 0$

- Truth value: **True (1)**
- Example: $x = -1$ (or any negative integer)

3. $\exists x \in \mathbb{N}, x \leq 0$

- Truth value: **False (0)**
- All natural numbers are positive (or ≥ 1 depending on convention)

Quantifiers - Practice Examples

Determine the truth value of each statement:

1. $\forall x \in \mathbb{N}, x \geq 1$

- Truth value: **True (1)** - all natural numbers are ≥ 1

2. $\exists x \in \mathbb{N}, x + 1 = 3$

- Truth value: **True (1)** - when $x = 2$

3. $\forall x \in \mathbb{R}, 2x - 6 = 2(x - 3)$

- Truth value: **True (1)** - this is an algebraic identity

4. $\forall x \in \mathbb{N}, x \leq 10$

- Truth value: **False (0)** - counterexample: $x = 11$

Universal Quantifier Examples:

- $\forall i \in \text{array}, \text{array}[i] \geq 0$
 - “All elements in the array are non-negative”
- $\forall x \in \text{List}, \text{isSorted}(x)$
 - “All sublists are sorted”

Existential Quantifier Examples:

- $\exists x \in \text{database}, x.\text{id} = 42$
 - “There exists a record with ID 42”
- $\exists \text{path} \in \text{graph}, \text{connects}(A, B)$
 - “There exists a path from node A to node B”

Negating Quantifiers

Important Rules:

Negation Rules

$$\neg(\forall x, P(x)) \equiv \exists x, \neg P(x)$$

$$\neg(\exists x, P(x)) \equiv \forall x, \neg P(x)$$

Examples:

1. Negate: “All students passed the exam”

- Original: $\forall s, \text{passed}(s)$
- Negation: $\exists s, \neg \text{passed}(s)$ - “At least one student failed”

Negating Quantifiers - Cont.

2. Negate: “There exists a prime even number greater than 2”

- Original: $\exists p > 2, \text{isPrime}(p) \wedge \text{isEven}(p)$
- Negation: $\forall p > 2, \neg(\text{isPrime}(p) \wedge \text{isEven}(p))$
- “All primes greater than 2 are not even” (which is true!)

Mixed Quantifiers

Statements can have multiple quantifiers:

Example 1: $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y > x$

- “For every real number x , there exists a real number y greater than x ”
- Truth value: **True** - we can always find a larger number

Example 2: $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y = y$

- “There exists a real number x such that for all y , $x + y = y$ ”
- Truth value: **True** - when $x = 0$ (additive identity)

Note: Order matters! $\forall x \exists y$ is different from $\exists y \forall x$

Summary: Quantifiers

Type	Symbol	Meaning
Universal	$\forall$	For all / Every
Existential	$\exists$	There exists / Some

Key Points:

- $\forall$ requires the statement to be true for **all** elements
- $\exists$ requires the statement to be true for **at least one** element
- Negation swaps quantifiers: $\neg\forall \equiv \exists\neg$ and $\neg\exists \equiv \forall\neg$
- Domain specification is crucial!

Practice: Try to express everyday statements using quantifiers!